

Exploratory Data Analysis (EDA) Report: Game Success Prediction

Introduction

Exploratory Data Analysis (EDA) is a critical step in understanding a dataset before applying machine learning models. In this project, EDA is performed on a video game dataset (`all_games.csv`) to analyze patterns related to game success using review scores and other attributes. The analysis helps identify trends, relationships, and insights that contribute to predicting whether a game is a **Hit** or **Flop**.

Project Objective

The main objective of this project is to analyze video game data and predict game success based on **Meta Score** and related features. The project also includes clustering and visualization to better understand game performance across platforms.

Dataset Description

Dataset Name: `all_games.csv`

Variables in the Dataset

- **Numerical Variables:**
 - Meta Score
 - User Review
 - **Categorical Variables:**
 - Game Name
 - Platform
 - **Derived Variable:**
 - Hit/Flop (Hit if Meta Score ≥ 75 , else Flop)
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Project Structure

1. Understand the Problem and Data
2. Import and Inspect the Data
3. Data Cleaning and Missing Value Handling
4. Exploratory Data Analysis
5. Data Visualization

6. Correlation Analysis
 7. Clustering Analysis
 8. Model Training
 9. Dash board
 10. Insights and Conclusion
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1. Understand the Problem and the Data

The gaming industry relies heavily on reviews and ratings to measure a game's success. By analyzing historical game data, this project aims to identify patterns that distinguish successful games from unsuccessful ones.

2. Import and Inspect the Data

File Used: data_loader.py

The dataset is loaded using Pandas. Initial inspection includes: - Viewing the first few records using `head()` - Understanding dataset structure using `info()` - Listing column names and data types

This step ensures familiarity with the dataset before further processing.

```
   index  name  platform  release_date  summary  meta_score  user_review
0      0  The Legend of Zelda: Ocarina of Time  Nintendo 64  November 23, 1998  As a young boy, Link is tricked by Ganondorf, ...    99      9.1
1      1  Tony Hawk's Pro Skater 2  PlayStation  September 20, 2000  As most major publishers' development efforts ...    98      7.4
2      2  Grand Theft Auto IV  PlayStation 3  April 29, 2008  [Metacritic's 2008 PS3 Game of the Year; Also ...    98      7.7
3      3  SoulCalibur  Dreamcast  September 8, 1999  This is a tale of souls and swords, transcendi...    98      8.4
4      4  Grand Theft Auto IV  Xbox 360  April 29, 2008  [Metacritic's 2008 Xbox 360 Game of the Year; ...    98      7.9
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 18800 entries, 0 to 18799
```

```
Data columns (total 7 columns):
#   Column      Non-Null Count  Dtype
---  -
0   index      18800 non-null    int64
1   name       18800 non-null    object
2   platform   18800 non-null    object
3   release_date 18800 non-null    object
4   summary    18686 non-null    object
5   meta_score  18800 non-null    int64
6   user_review 18800 non-null    object
dtypes: int64(2), object(5)
memory usage: 1.0+ MB
```

3. Data Cleaning and Preprocessing

File Used: preprocessing.py

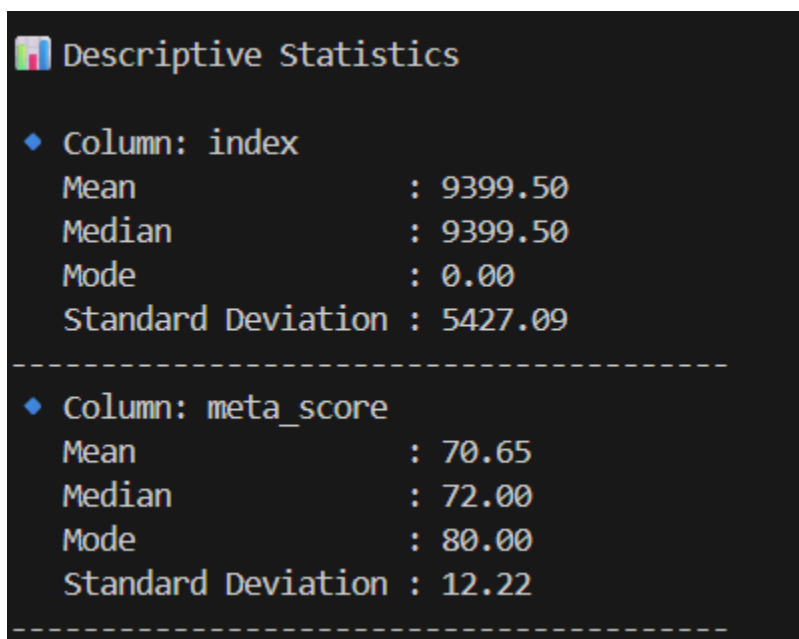
The following cleaning steps were performed: - Handling missing values in numerical columns - Removing unnecessary or invalid records - Ensuring correct data types for modeling

After preprocessing, the dataset is clean and ready for analysis and modeling.

4. Exploratory Data Analysis

Descriptive statistics were computed to understand the distribution of numerical features: - Mean - Median - Standard Deviation - Minimum and Maximum values

These statistics provide insights into overall game rating trends and score variability.



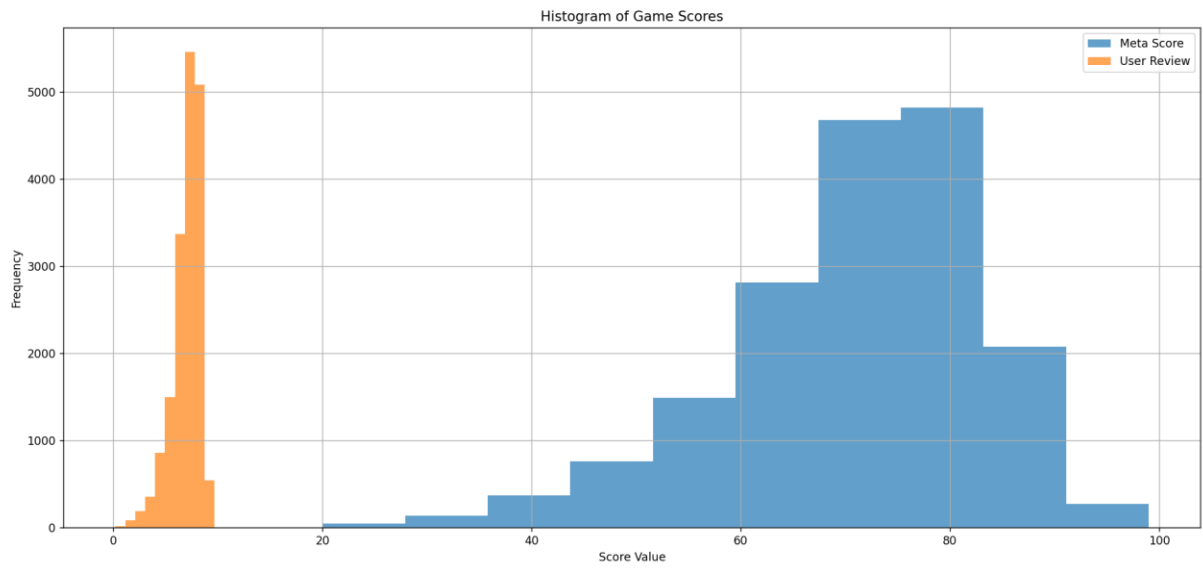
5. Data Visualization

Files Used: histogram.py, kde_plot.py, visualization.py

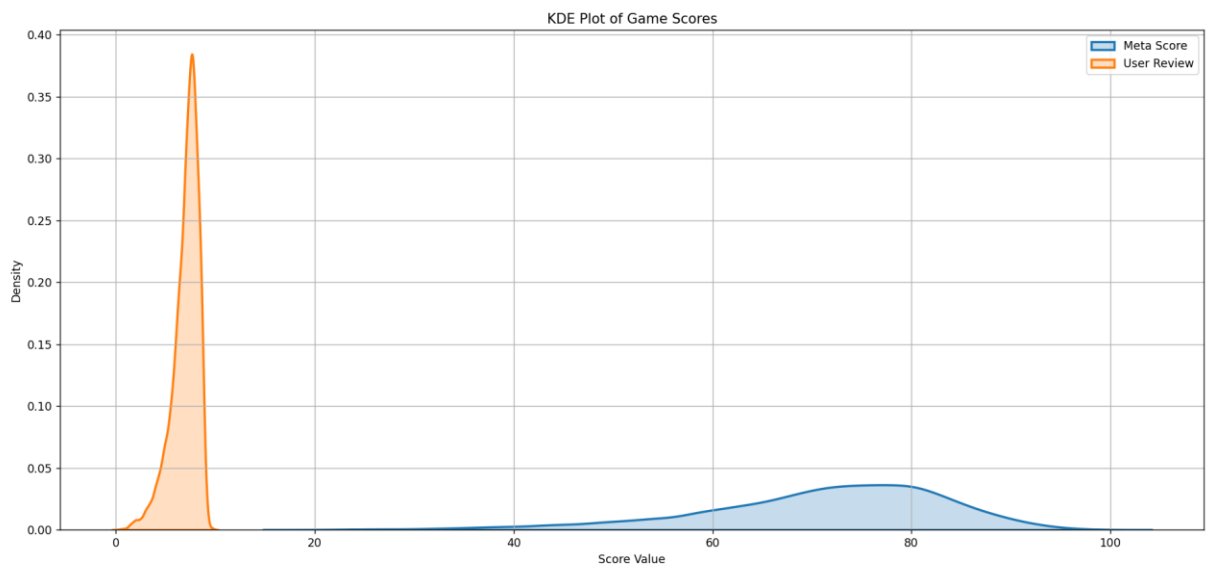
The following visualizations were created: - **Histogram:** Distribution of Meta Scores - **KDE Plot:** Density estimation of score distribution - **Platform-wise Score Plot:** Average Meta Score across gaming platforms

These plots help visualize how game ratings vary and which platforms perform better.

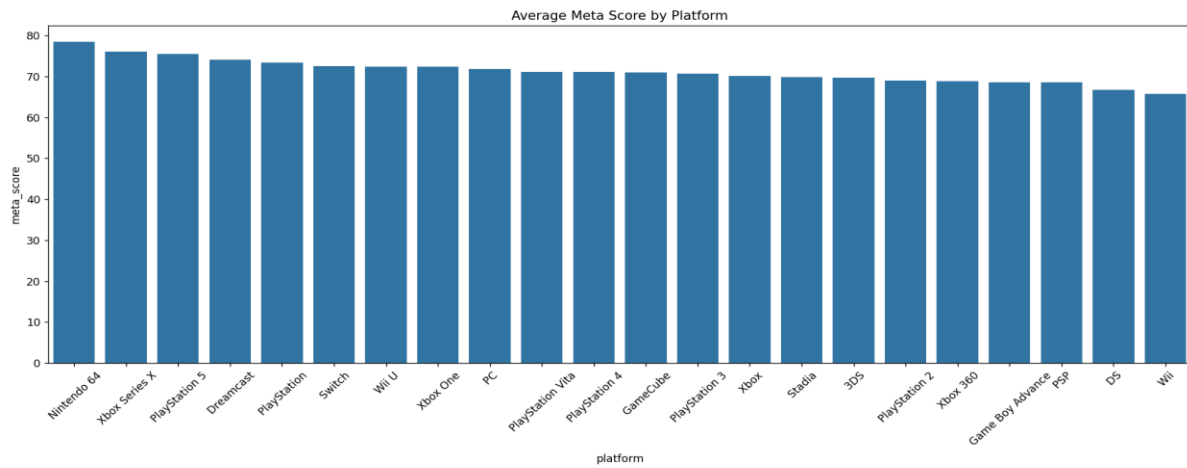
Histogram plot:



Kde plot:



Visualization plot:

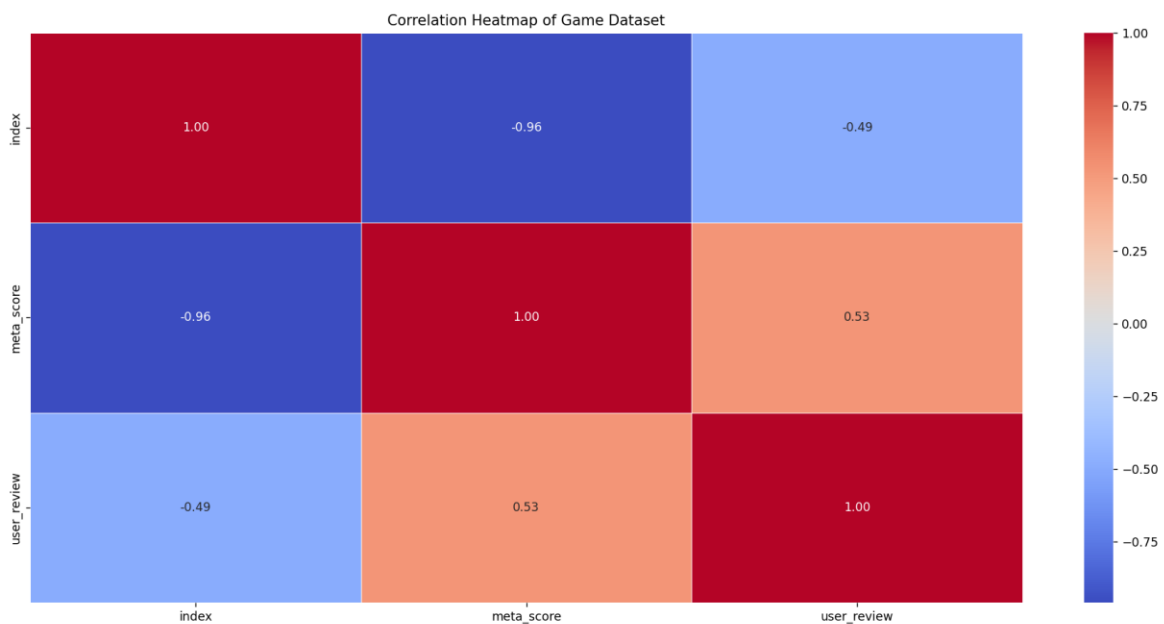


6. Correlation Analysis

File Used: correlation.py

A correlation heatmap was generated to analyze relationships between numerical variables. This helps identify features that strongly influence game success.

Correlation plot:

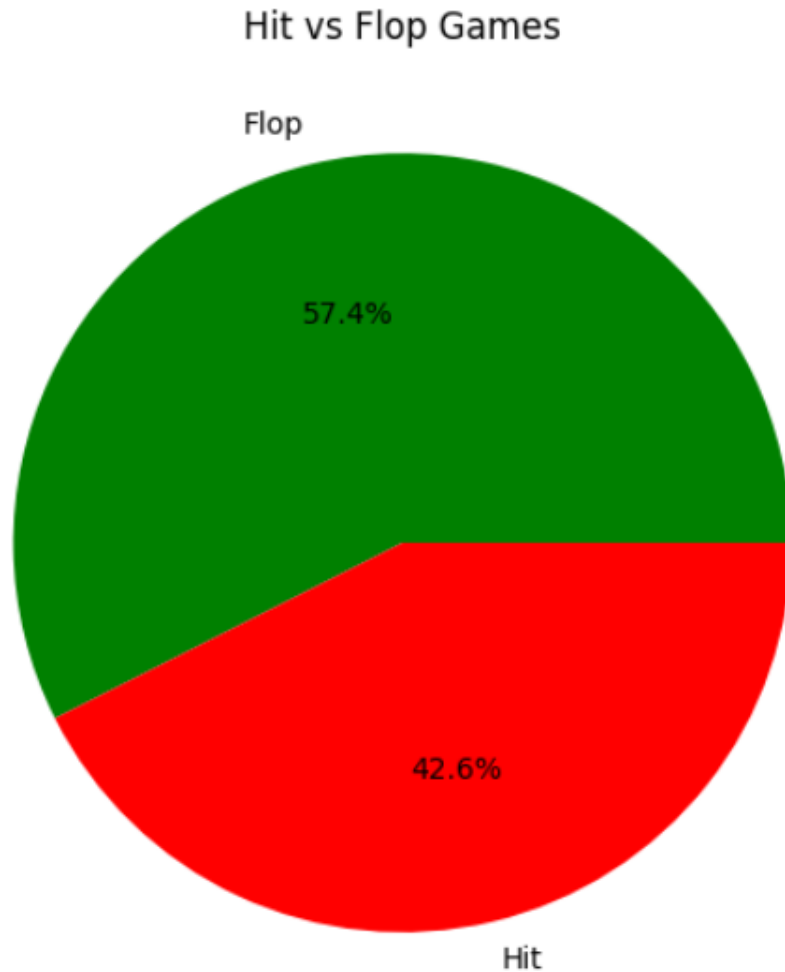


7. Hit / Flop Classification

A new column `hit_flop` was created: - **Hit**: Meta Score ≥ 75 - **Flop**: Meta Score < 75

This classification simplifies success prediction and supports supervised learning.

Hit / Flop:

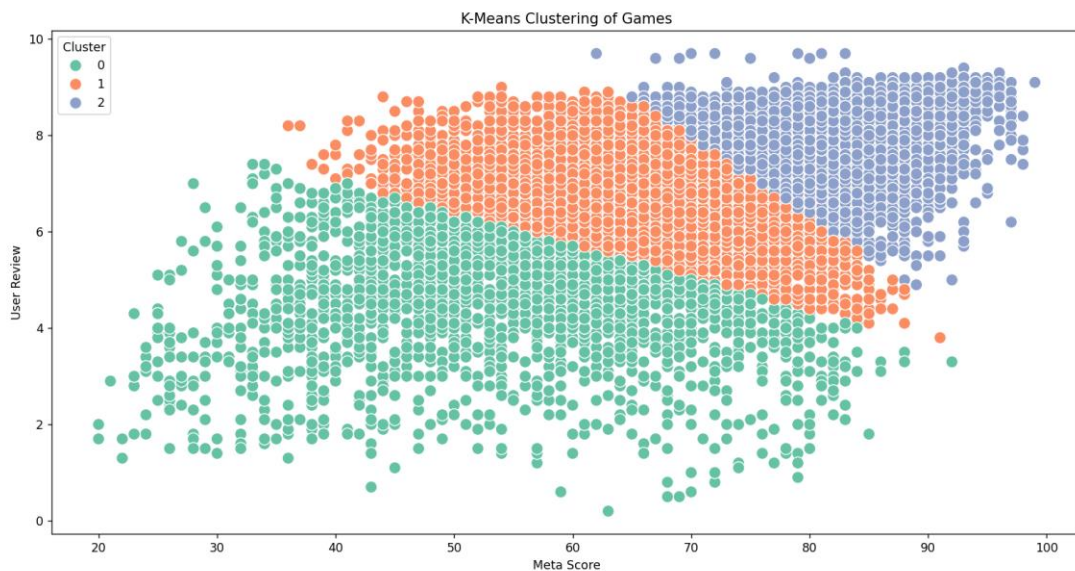


8. Clustering Analysis

File Used: clustering.py

K-Means clustering was applied with $k = 3$ to segment games into performance-based clusters. This unsupervised approach reveals hidden groupings among games with similar characteristics.

Kmean clustering:



9. Model Training

File Used: model.py

A Random Forest Regressor was trained to predict Meta Scores. Model performance was evaluated using: - Mean Absolute Error (MAE) - R^2 Score

The results showed high accuracy, indicating strong predictive capability.

```
Model Performance (Regression)
MAE: 0.0039454787234042416
R2 Score: 0.9999861812030663
```

10. Dash board

File Used: dashboard.py

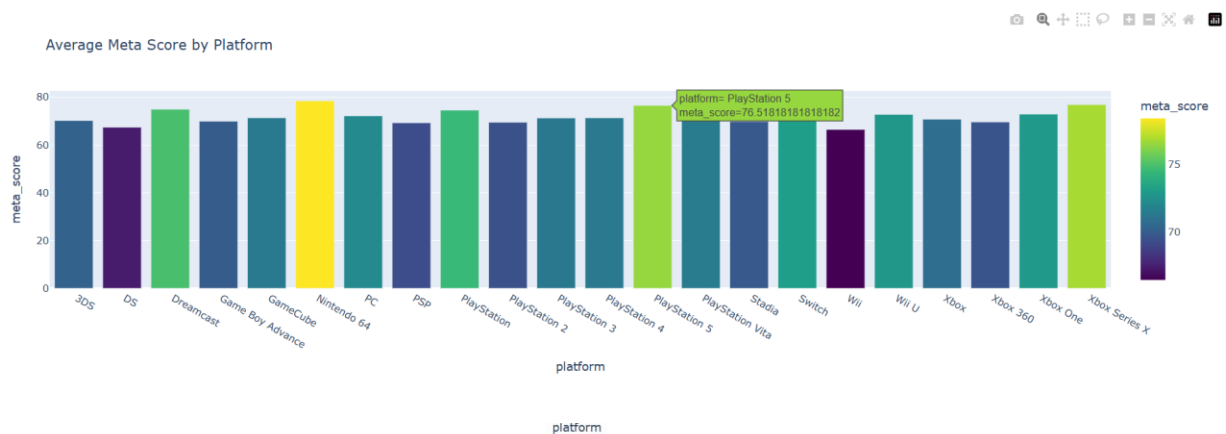
Dash enables the creation of interactive reports by combining Python, Plotly, and HTML components.

In this project, Dash is used to present Exploratory Data Analysis (EDA), descriptive statistics,

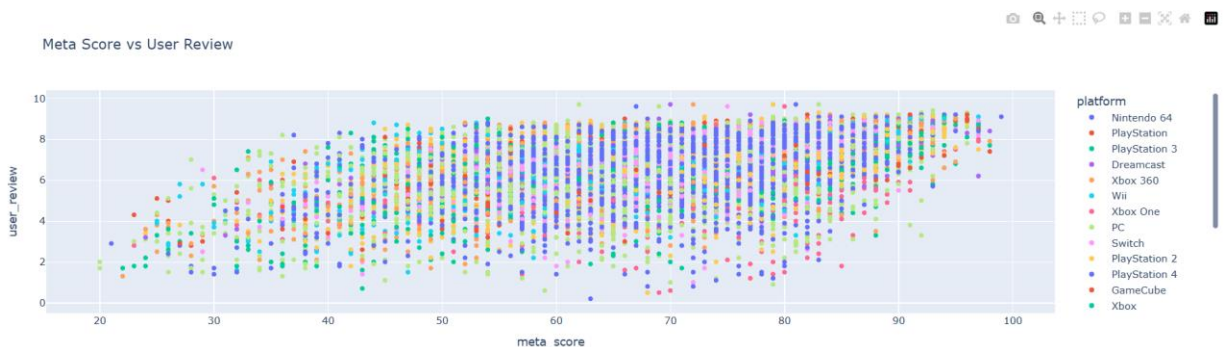
and machine learning model results in a single interactive interface.

Game Success Dashboard

Platform-wise Average Meta Score



Meta Score vs User Review



Conclusion

This project successfully performed end-to-end Exploratory Data Analysis on a video game dataset. Through data cleaning, visualization, correlation analysis, clustering, and machine learning, meaningful insights into game success were obtained. The study demonstrates how data science techniques can be applied to predict game performance and support decision-making in the gaming industry.